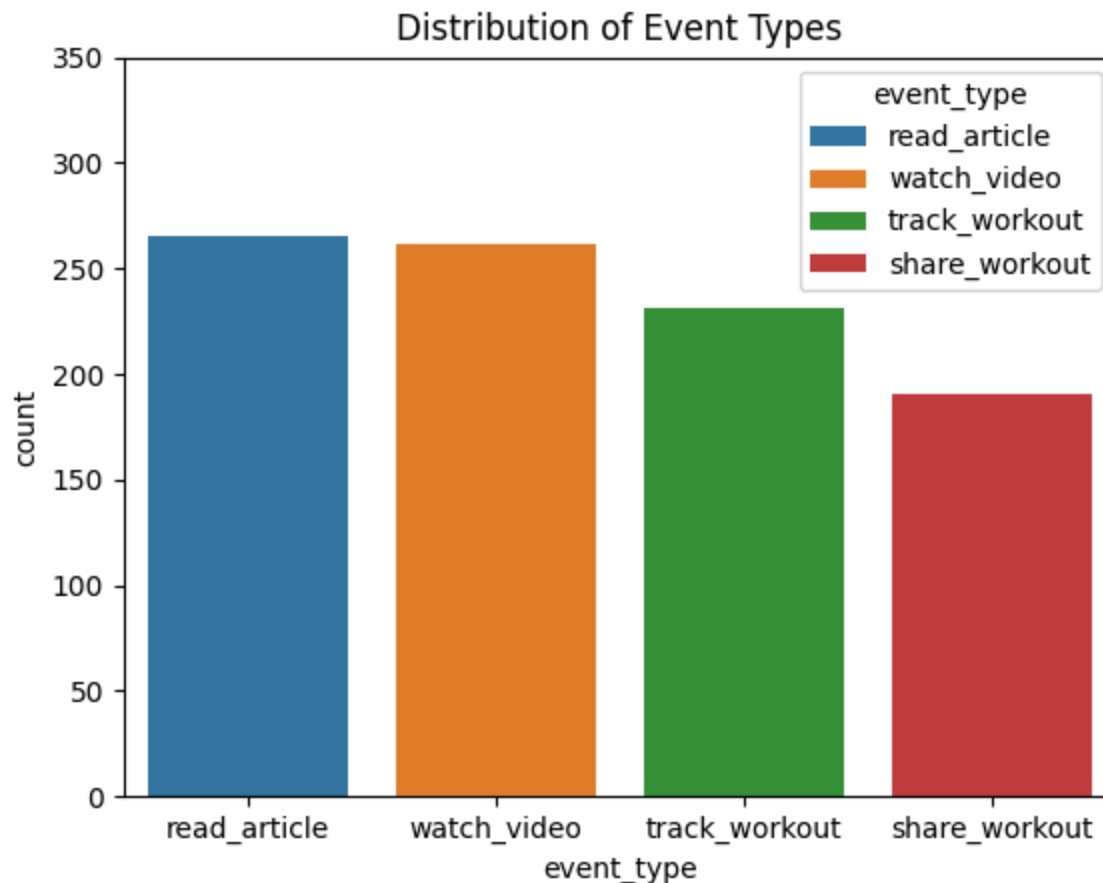


Churn Analysis for Fit.Ly

Introduction

Fit.Ly is experiencing elevated customer churn and asked me to identify the root causes and provide actionable recommendations to improve retention. Fit.Ly is a consumer fitness app offering workout articles, instructional videos, workout tracking, and the ability to share workouts privately or on social media.



I was provided three datasets for this analysis:

- 1) User account information
- 2) Customer support tickets
- 3) User activity logs

Data Validation

All three datasets were validated to ensure consistent identifiers, standardized labels, correct data types, and reliable values. This preparation ensured accurate churn and retention analysis.

Account_Info Table

customer_id: Removed leading “C” and converted to integers (400 unique values). Used to merge with user_ids in other datasets.

Email: 400 unique values, no duplicates.

State: 50 unique values, representing all U.S. states.

Plan: 4 unique values, as expected. Converted to category type.

Plan_list_price: Converted to float; values ranged from 0 to 148 USD.

Churn: Standardized by converting "Y" to "Churned" and filling nulls with "Not Churned". Converted to category type.

Customer Support Tickets Table

Ticket_time: Converted to datetime64[ns] for datetime analysis.

User_id: Aligned with identifiers in the other datasets for merging.

Channel: 4 unique values; “-” recoded as “Unknown”. Converted to category type.

Topic: 4 unique values, converted to category type.

Resolution_time_hours: Valid float values from 0.52 to 32.46 hours.

State: Contained only 0 and 1 values with no documented meaning or explanation from Fit.Ly Product Manager or Lead Engineer; removed.

User Activity Logs Table

Event_time: Converted to datetime64[ns] for datetime analysis.

User_id: Consistent with identifiers across all tables.

Event_type: 4 unique values, converted to category.

Data Deletion Requests

A total of **43 customers** requested data deletion, resulting in **184 records** being removed across all datasets before the analysis.

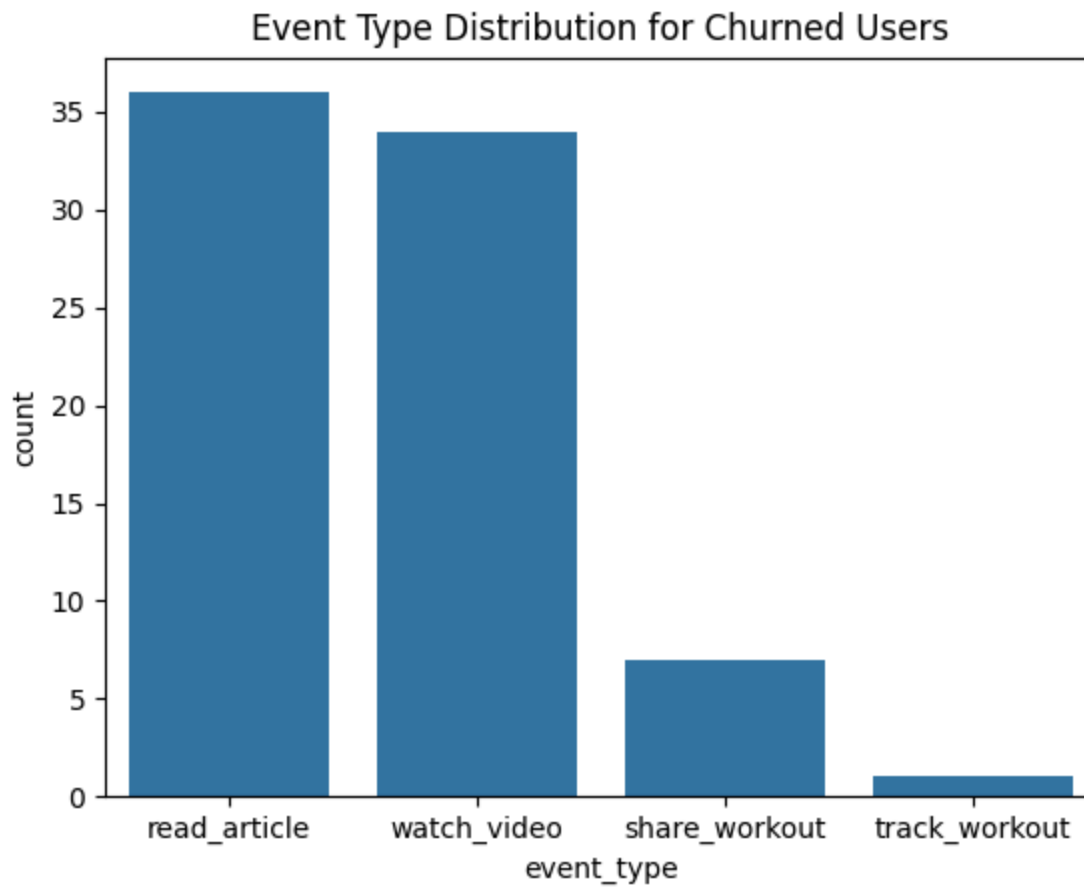
This was necessary for legal compliance and upholding company standards on ethics and business practices.

Exploratory Analysis

After cleaning and organizing the data, I found two key insights about churned users.

1. Churned customers were overwhelmingly passive users

Churned users primarily consumed content rather than engaging with Fit.Ly's core fitness features:



46% reading articles

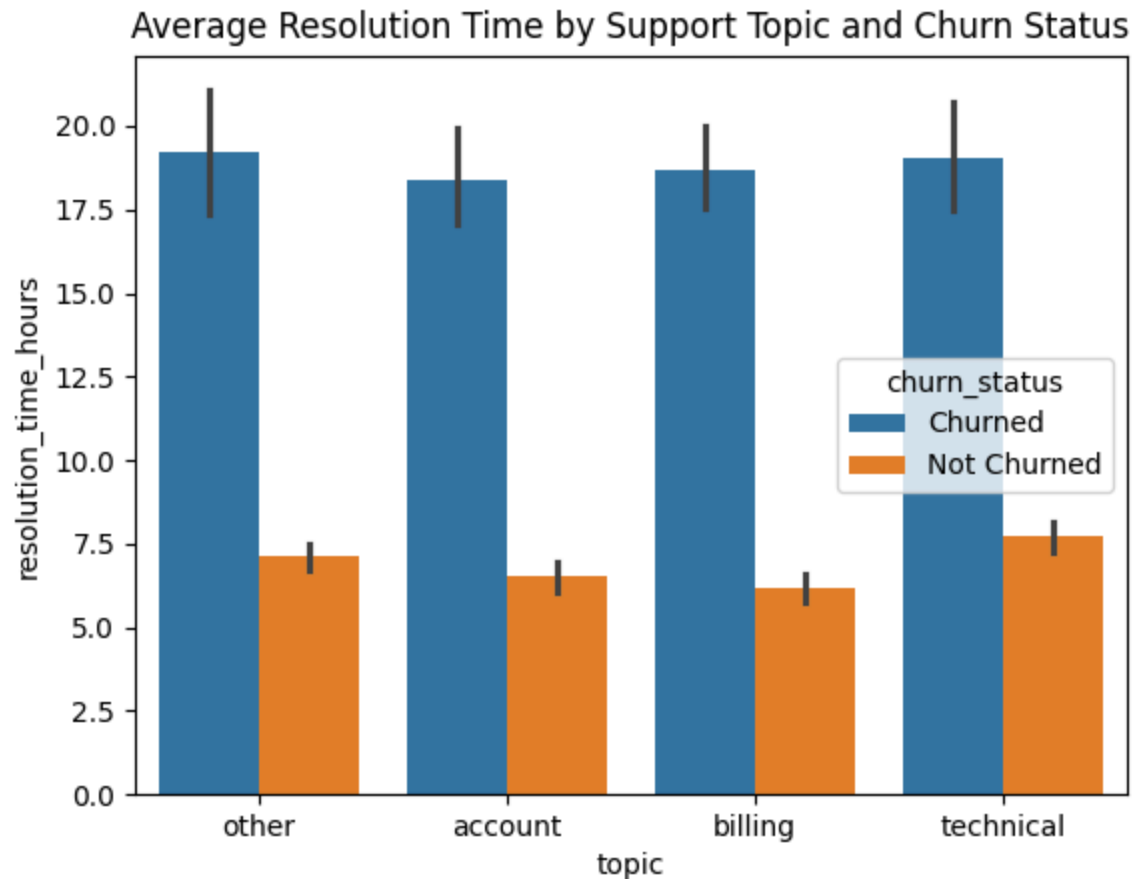
44% watching videos

1% tracking workouts

Overall, ~90% of churned users were passive consumers.

2. Passive behavior alone did NOT cause churn

The key differentiator between retained and churned users was customer support resolution time:



Retained users waited **6.87 hours** on average

Churned users waited **18.82 hours** on average

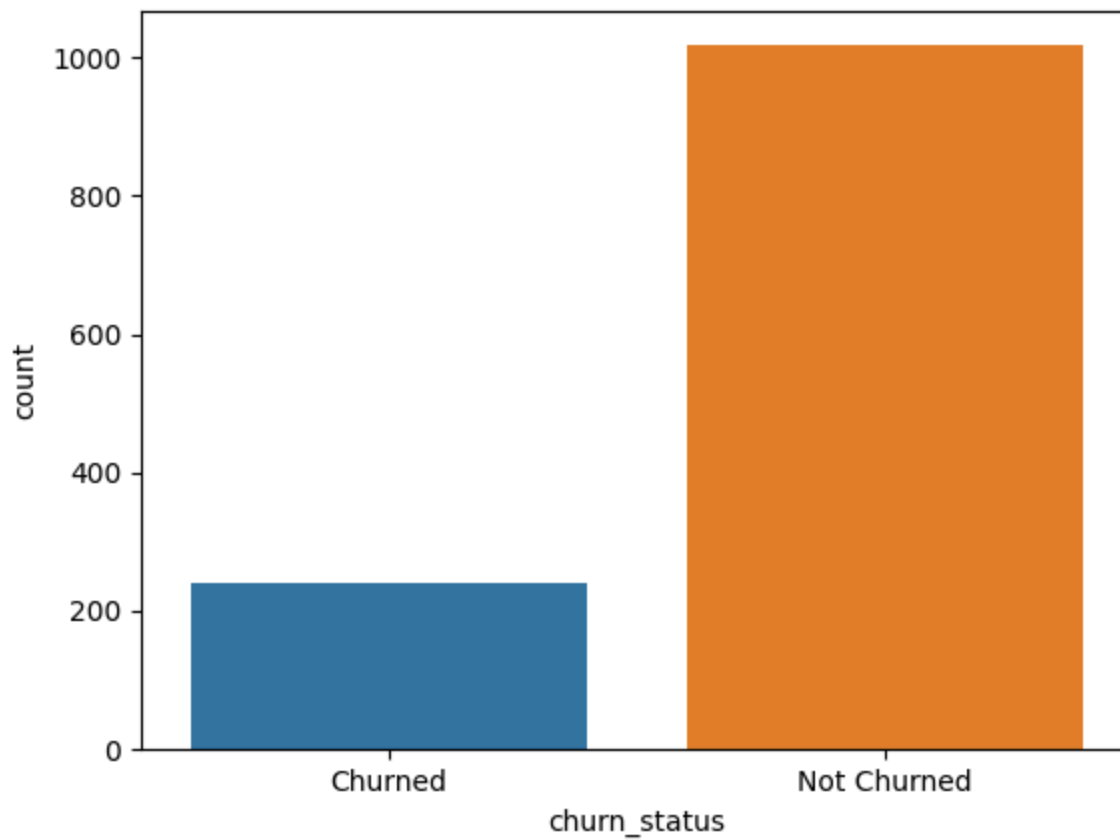
This 2.5× longer delay suggests that slow support pushed users toward alternative apps.

Key Business Metric: Customer Churn Rate

Fit.Ly's primary business metric is customer churn rate — the percentage of users who discontinue the service. Churn is a critical indicator of product health because improvements in retention compound into higher recurring revenue and customer lifetime value.

Baseline:

Fit.Ly's current churn rate is 19% (almost 1/5). 240 users out of 1,257 left the app in the last two quarters.



Why it matters:

Reducing churn is significantly more cost-effective than acquiring new users and directly impacts subscription revenue.

Key driver of churn:

Customer support resolution time is the strongest predictor of churn. Churned users waited 18.82 hours, compared to 6.87 hours for retained users. Among passive users who received support in under 7 hours, only 2% churned.

How to track it:

Monitor churn monthly, segmented by user type and support experience. A dashboard combining churn rate, resolution time, and ticket volume will help identify when operational delays begin affecting retention.

Business decision supported:

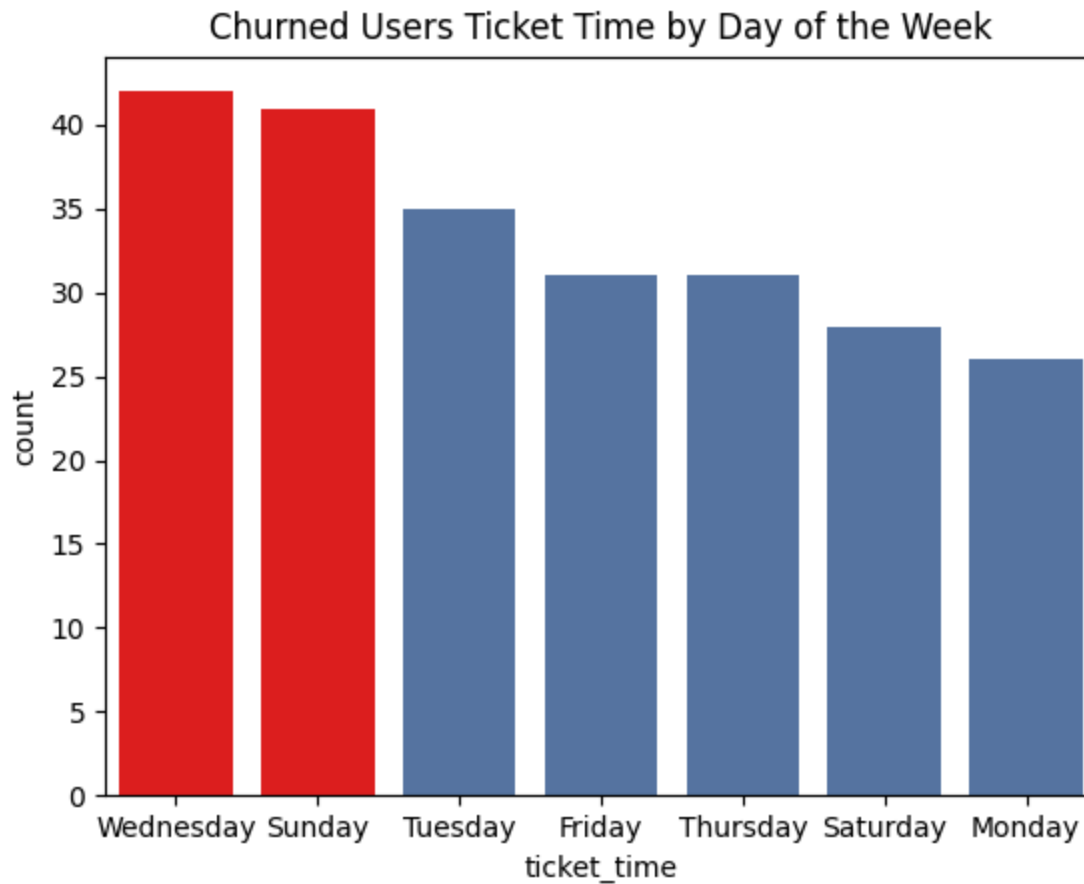
Tracking churn alongside support performance enables Fit.Ly to evaluate the impact of staffing changes, AI-assisted support tools, and

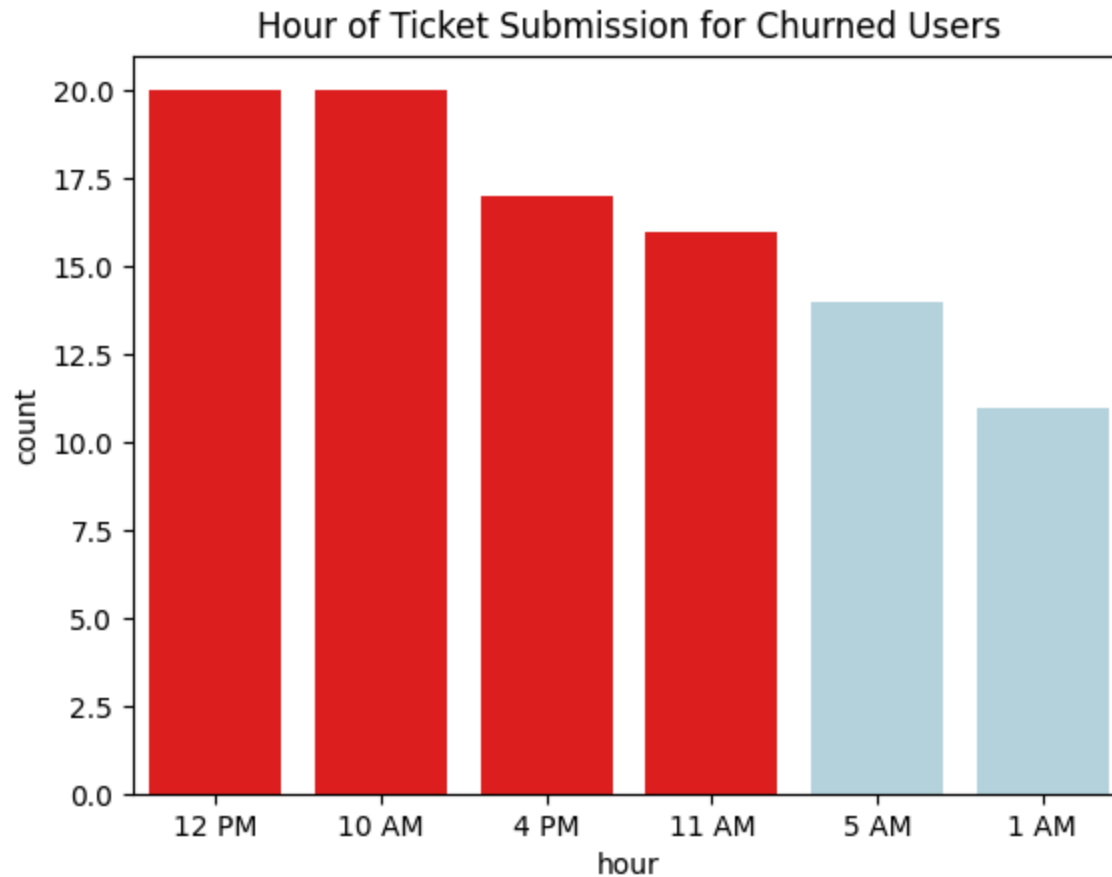
workflow improvements.

Customer Support Recommendation

Customer support resolution time is the most consistent driver of churn. Users who waited longer for help were significantly more likely to cancel, regardless of their subscription plan. This indicates that support delays are not merely correlated with churn — they actively contribute to it.

Operational patterns show that delays cluster in predictable windows. Support performance consistently deteriorates on Wednesdays and Sundays. The 10 AM to 4 PM window experiences the highest congestion, with resolution times rising well above average.





To address this, Fit.Ly should:

- Increase or retrain staffing on Wednesdays, Sundays, and during 10 AM – 4 PM PT
- Implement AI-based triage to instantly resolve simple issues and reduce queue buildup

Engagement Recommendation

Fit.Ly should also encourage passive users to engage with core fitness features. Articles and videos can include calls-to-action linking directly to workout tracking:

*“Enjoyed this office workouts article? **Try these quick workouts now!**”*

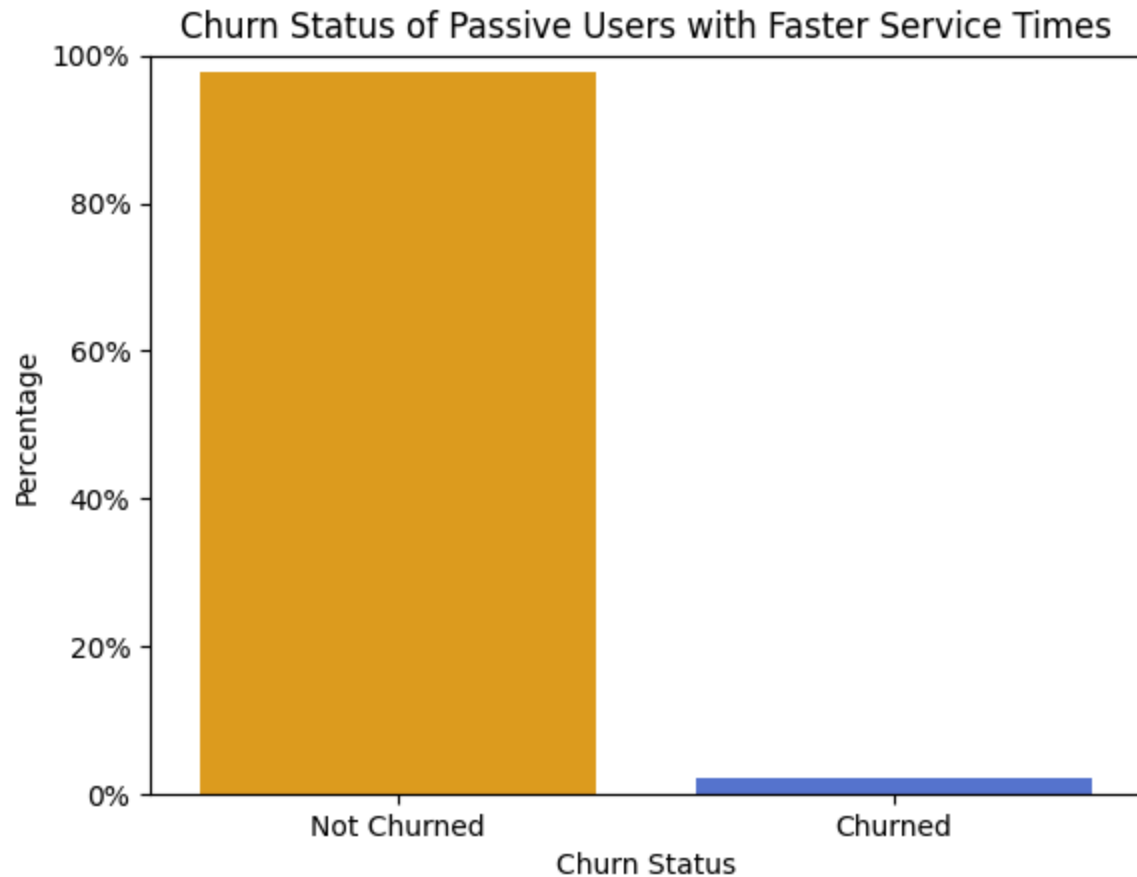
Since passive users make up most churned customers, increasing engagement could significantly improve retention.

Retention Test: Passive Users With Fast Support

To validate the impact of support speed, I analyzed passive users who:

- Only consumed articles and videos
- Received support resolution in under 7 hours

Among these users, only **2% churned** — meaning **98% stayed**.



This strongly confirms that customer support response time is the primary driver of churn, not passive use. If Fit.Ly reduces resolution times to the normal range (around 6.87 hours), the company can retain the vast majority of users who would otherwise leave.

Conclusion

Overall, this analysis shows that Fit.Ly's churn challenge is driven by a clear and solvable operational issue: slow customer support during predictable high-volume windows.

By improving staffing coverage, introducing AI-based triage, and encouraging deeper engagement with core fitness features, Fit.Ly can meaningfully reduce churn and strengthen long-term customer loyalty.

These recommendations directly address the root causes identified in the data and provide a practical roadmap for improving retention and overall product health.